Covariance and Correlation

:- Covariance is a statistical term that refers to a systematic relationship between two random variable. In which a change in the other reflect a change in one variable

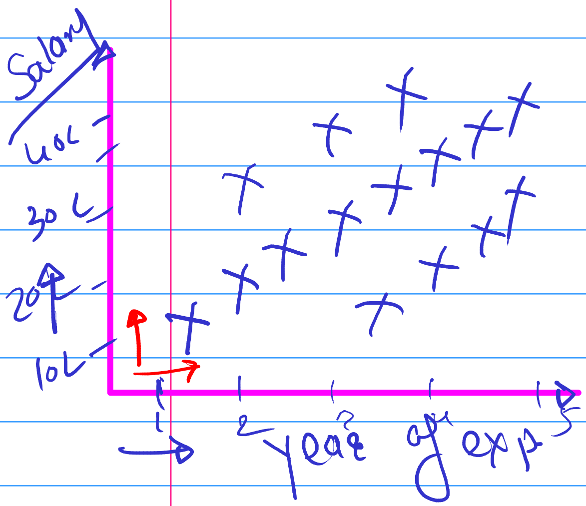
:→ The range of Covariance is between $-\infty$ to $+\infty$.

+ve | −ve

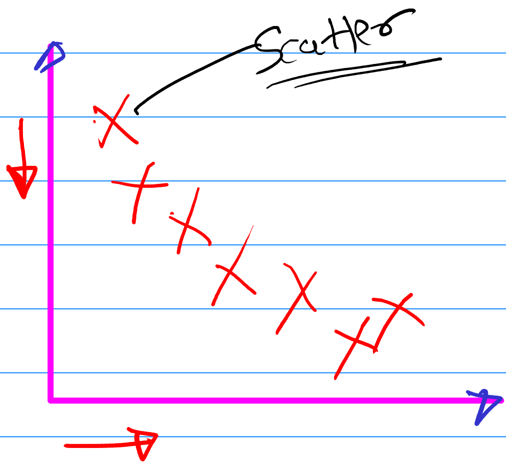
+30, +300 +3000 - - - - - · · ∞

− 30, − 300, − 3000 - - - $-\infty$

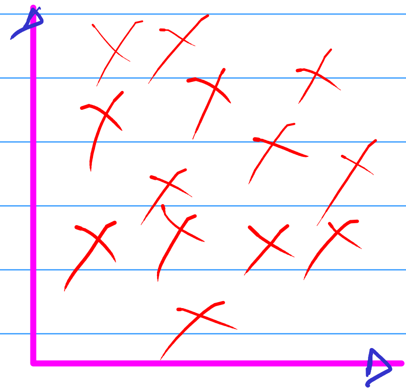
year of exp	Salary



+ve Covariance



−ve Covariance



Zero / No Covariance

↳ Disadvantage of Range $-\infty$ to $+\infty$

Correlation

If one variable increase dose the other variable increse or decrease.

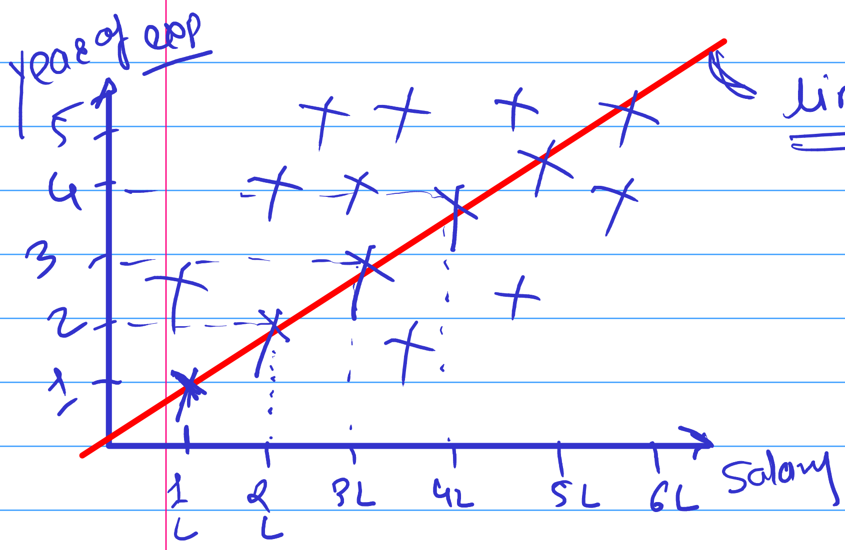
↑ ↑ | ↑ ↓

:- The range of Correlation is between −1 to +1

−100% 100%

① Pearson Correlation → Linear line

: It is the most common way of measuring a linear Correlation.



linear line

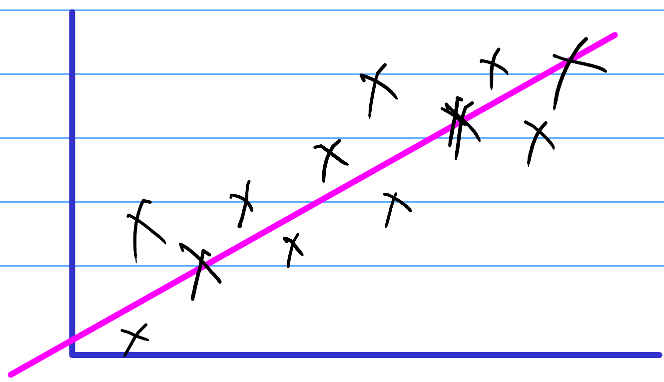
linier line

$y = mx + c$

Salary ↑ year ↑

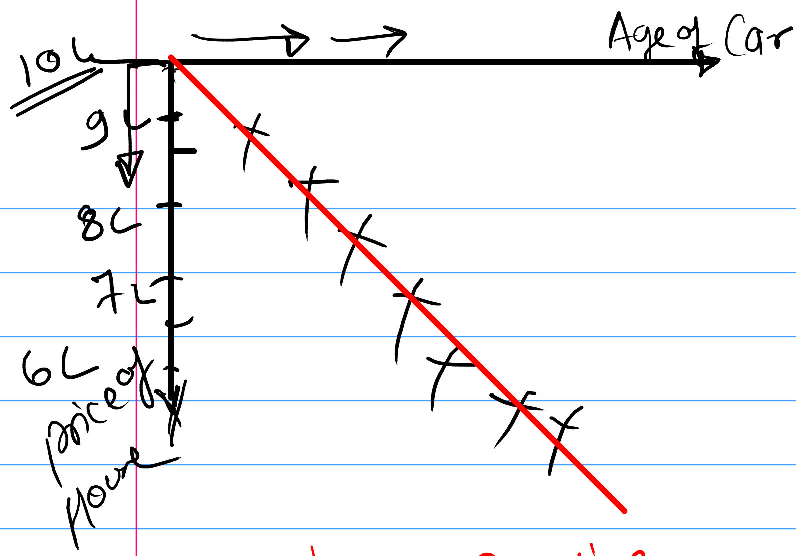
+ve Correlation

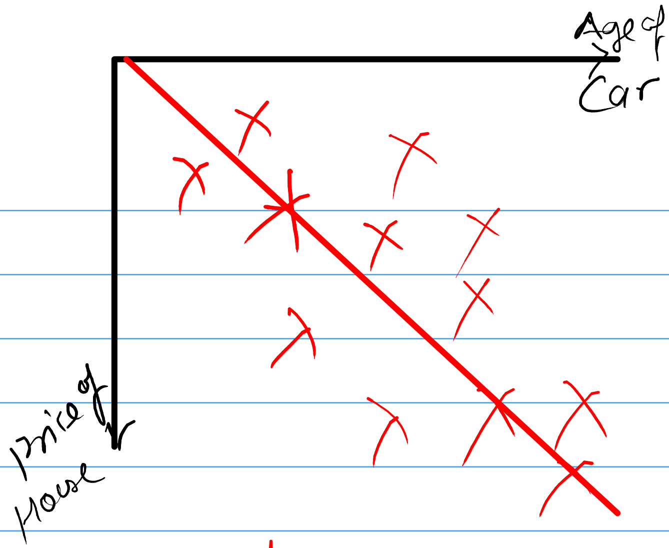
+100%



→ 0 to 1 Correlation

moderate positive

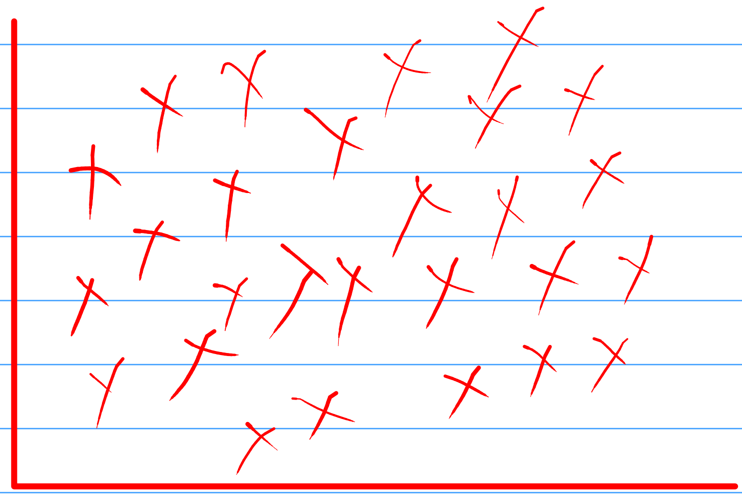




Strong negative -100% (-1)

moderate negative (-1 to 0)

Zero Correlation (0)



② Spearsman's Rank Correlation (-1 to +1)

:- use for the non-linearity

Types of Analysis

Columns

① Univariate → only one feature

② Bivariate → two feature

③ Multivariate → more than two

① Univariate Analysis

→ Categorical
→ Numerical

① Categorical Feature :→ Value_Counts()

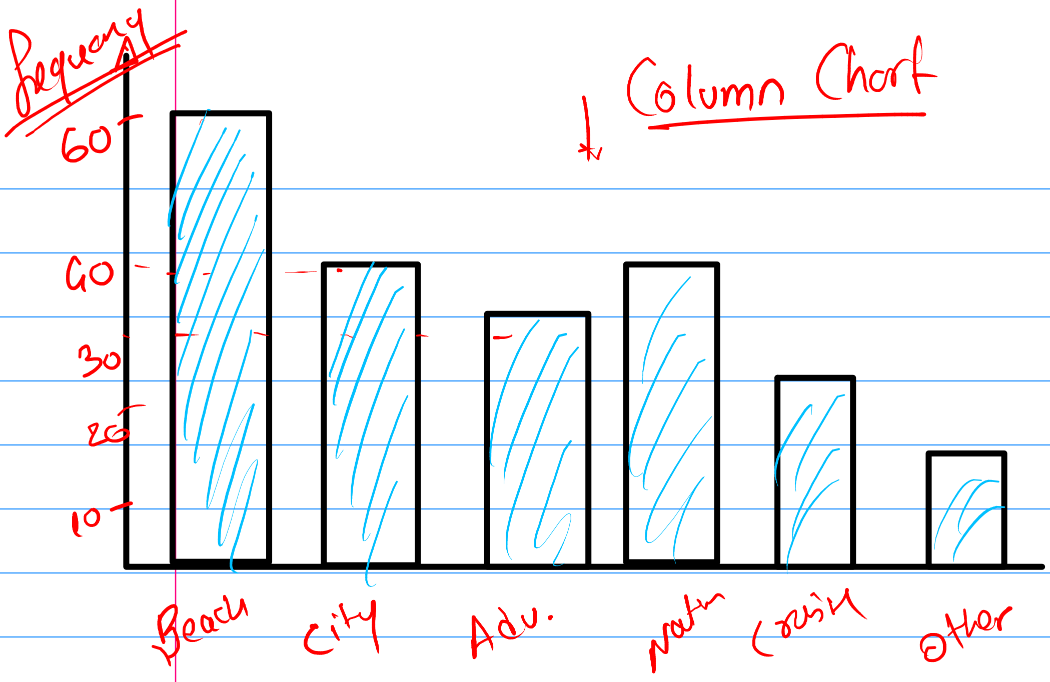
Frequency Distribution Table
count

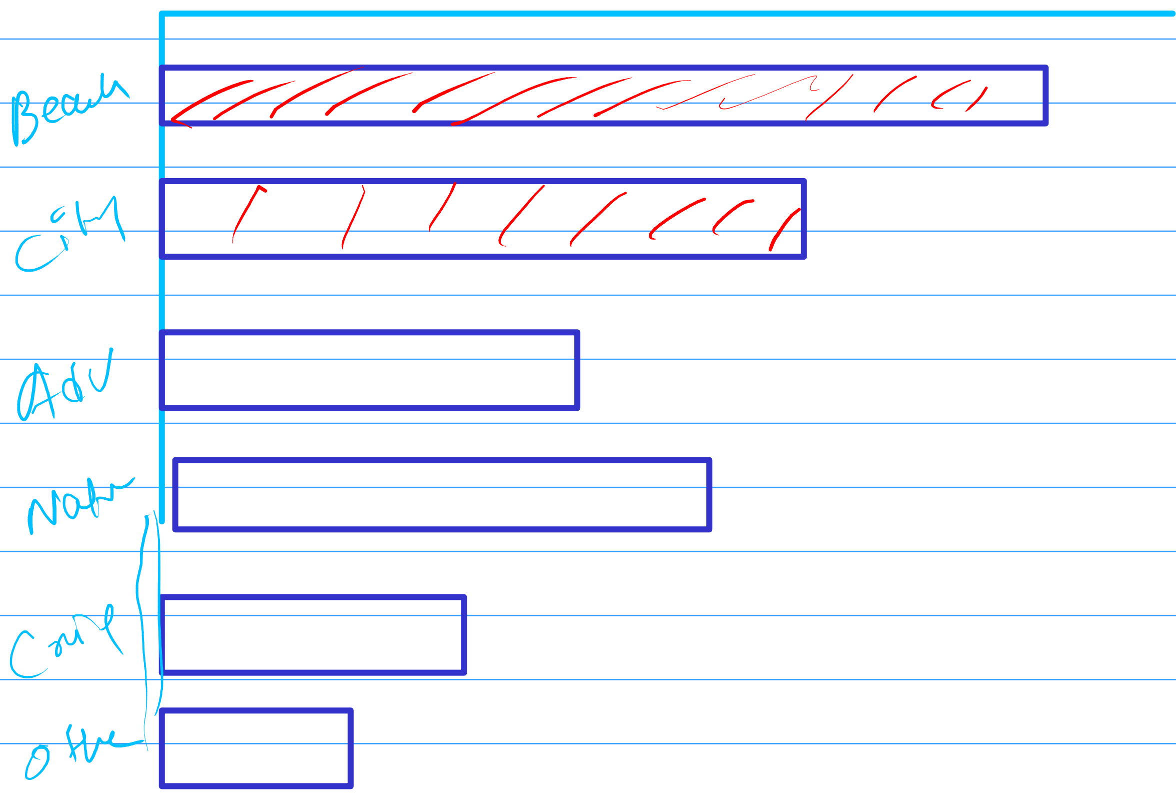
200 Student Vaccation Data.

Type of Vaccation	Frequency
Beach	60
city	40
Adventure	30
Nature	35
cruise	20
other	15

→ Bar chart / Column chart.

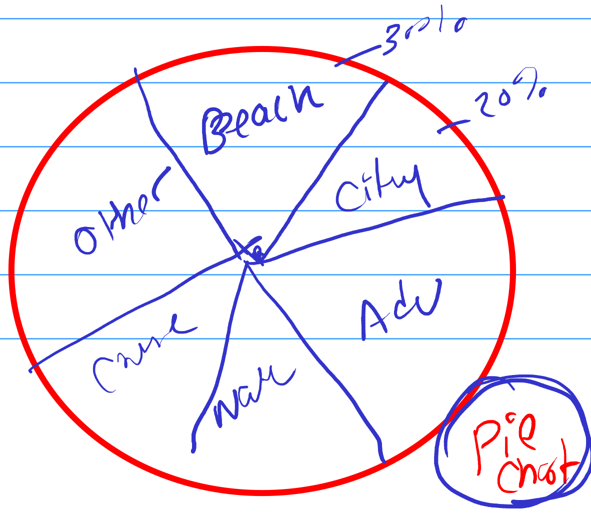
Column Chart



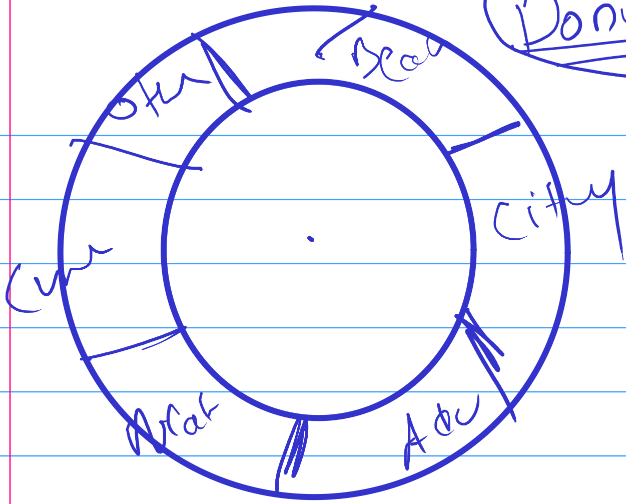


Distribution (Rate) → percentage → Relative frequency

	Freq	Relative Freq
Beach	60	60/200 = 0.3 = 30%
City	40	0.2 = 20%
Adv	30	0.15 = 15%
Nature	35	0.175 = 17.5%
Cruise	20	0.1 = 10%
Other	15	0.075 = 7.5%
		1 100%



Donut chart

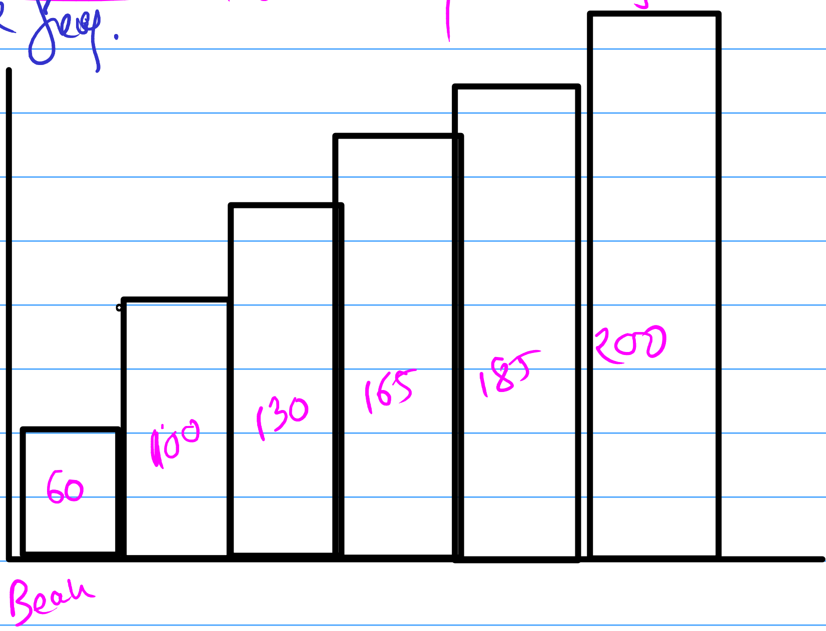


Cummulative frequency (Waterflow chart)

addition / Running total

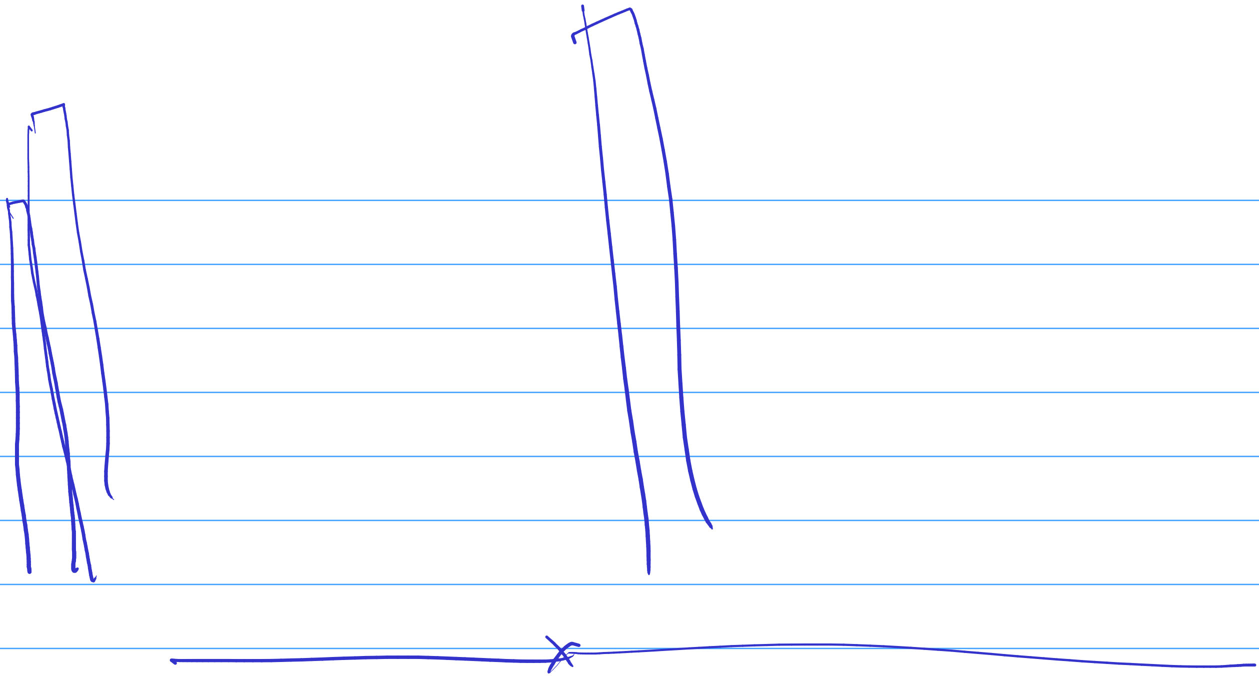
Cummulative freq.

Beach	60	60
City	40	100
Adv.	30	130
Nature	35	165
Cruise	20	185
Other	15	200



Month (profit)

J → 10L	J → 10L
F → 6L	A → 12L
M → 4L	S → 4L
A → 2L	O → 2L
M → 8L	N → 6L
J → 9L	D → 5L



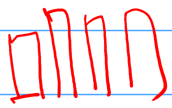
② Univariate Analysis – Numerical Column

Age

200 Student

8
9
12
15
28
25
9
8

→ Bar



8 9 10 11 12 13 - - - - - - - 30

Bins ⇒

Categorical conversion

Age group		
8-18	Teen	10
18-25	Adult	12
25-45	Sr. Adult	15
45-60	Senior	17
	old age	

Distribution (Numerical Data)

Histogram

Bins = ? = 30 → 3-4 bar

Bivariate Analysis

- Categorical vs cat
- Num vs Num
- Num vs Cat / Cat vs Num

① Cat vs Cat → Contengency Table / Cross Tab

Gender | exam
M | Pass
F | fail

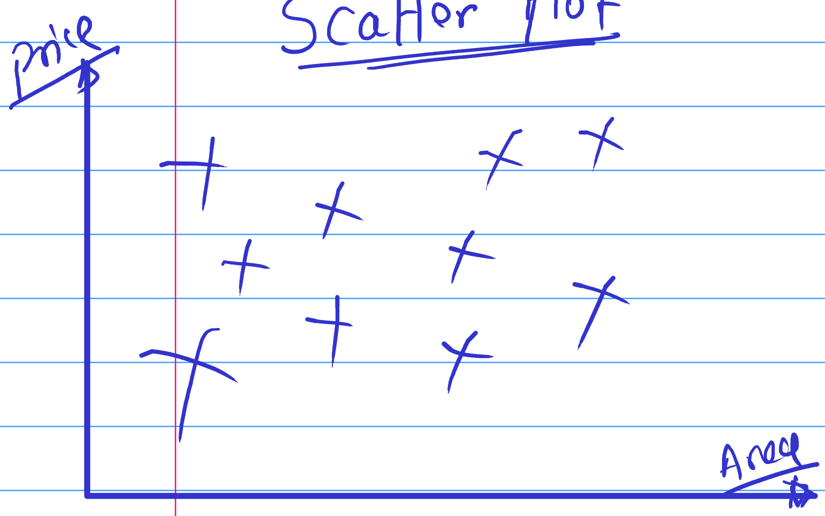
200 student

Result

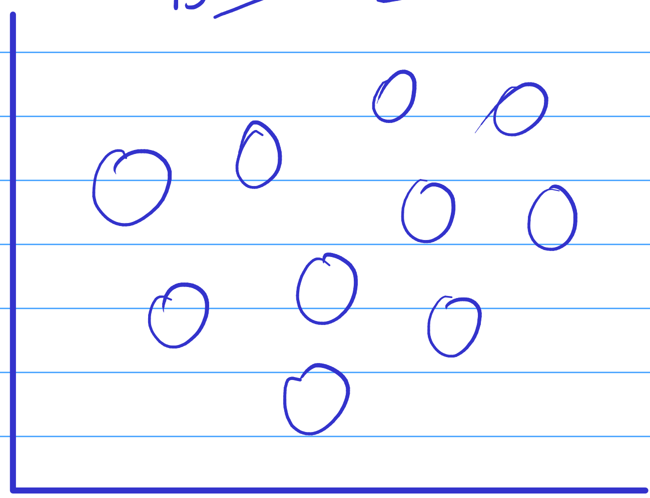
Gender	Pass	Fail
M	129	45
F	30	73

② Num Vs. Num

Scatter Plot



Bubble chart



③ Cat Vs Num / Num Vs Cat.

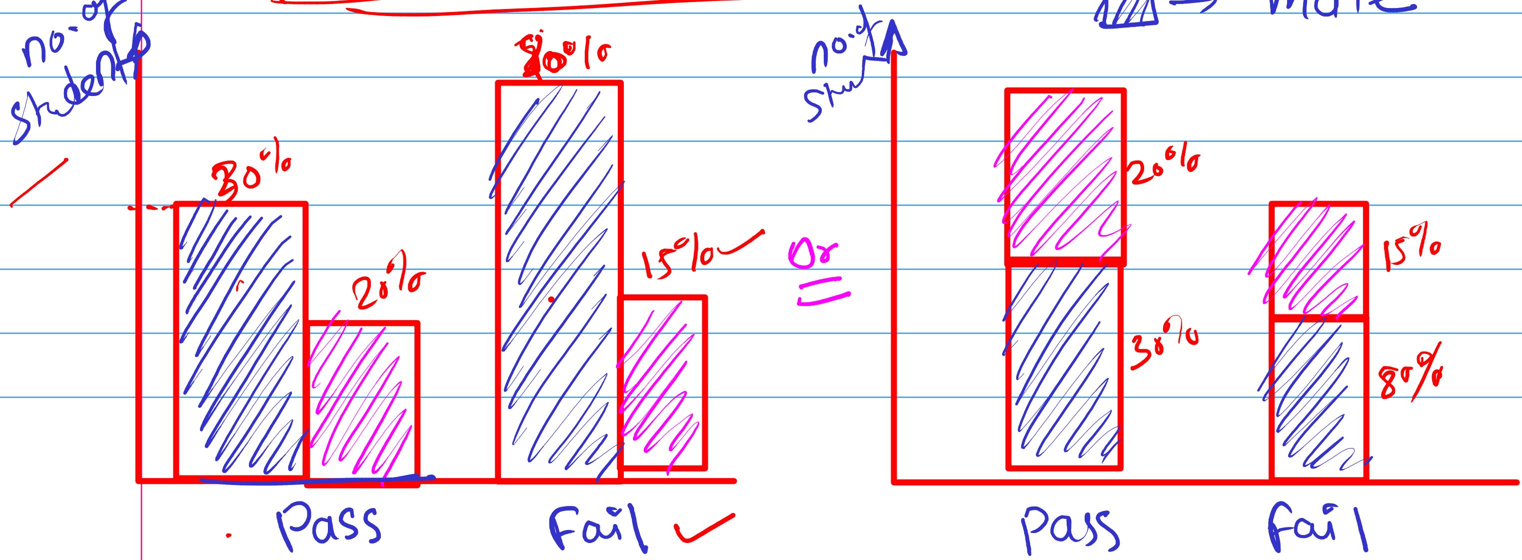
Age → Age_cat

	Teen	Adult	Sr	old age
Male				
Female				

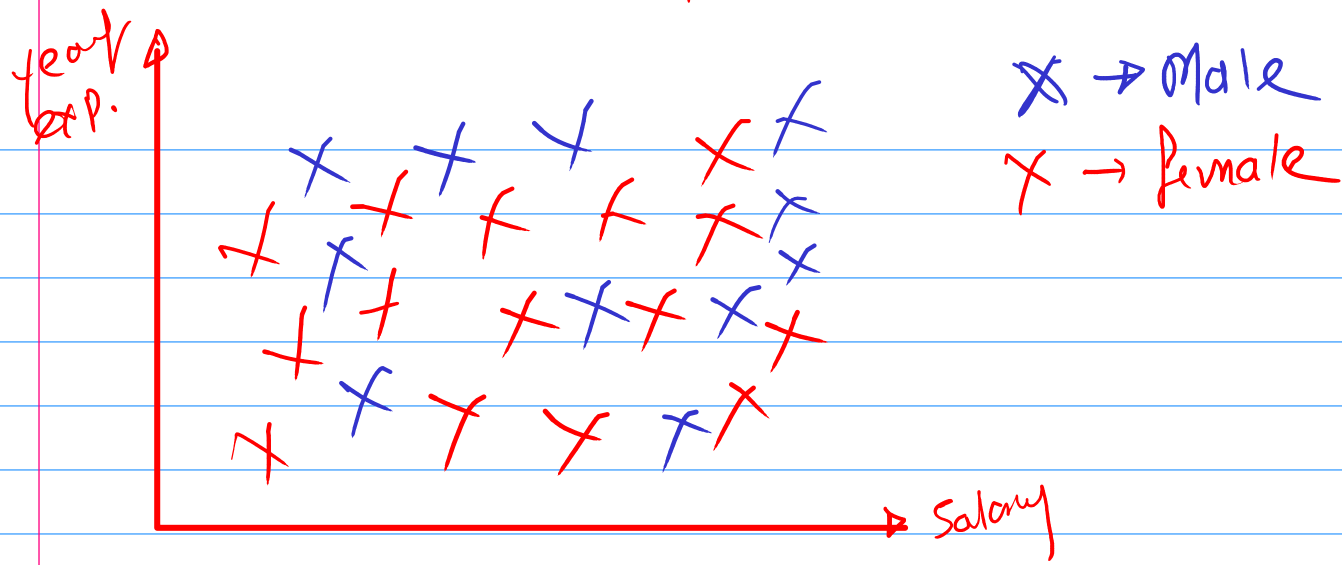
Contingency table

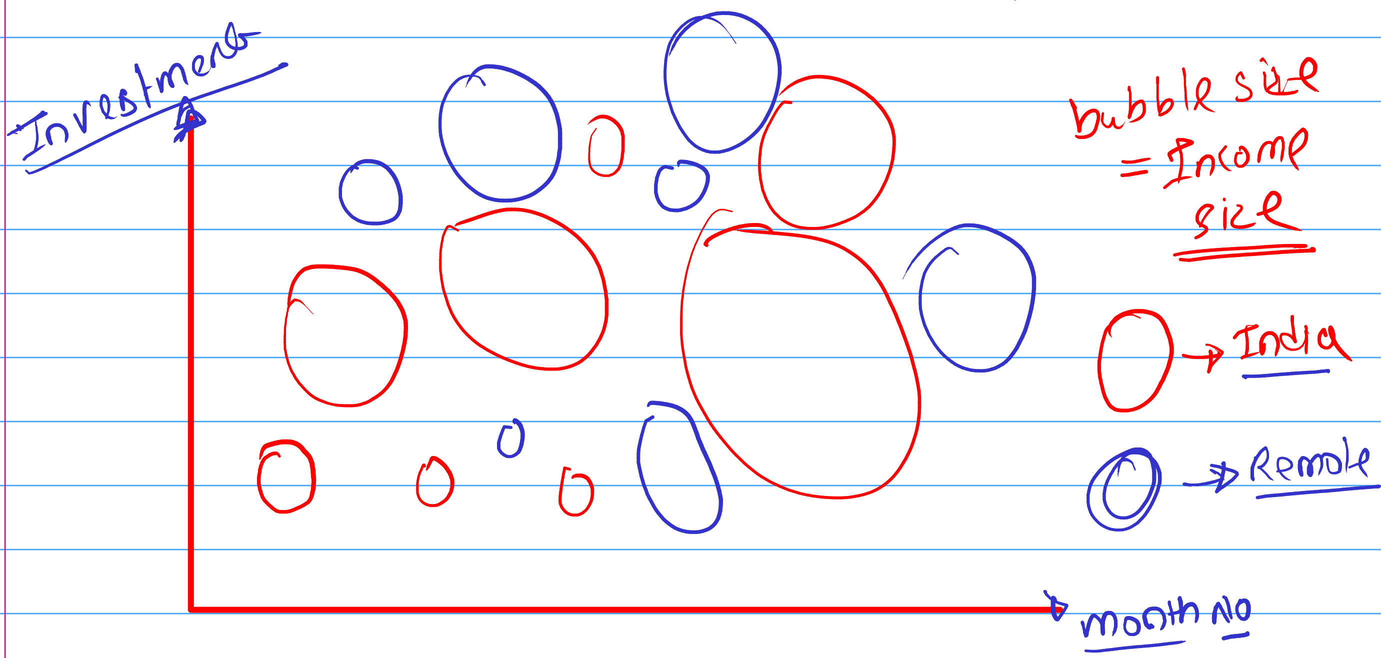
Multivariate Analysis

legend → Female
→ Male



num + Num + cat


year exp.
Salary
X → Male
X → Female


Investment
month No
bubble size = Income size
→ India
→ Remote